Combined executive report: Civilingenjör i AI och maskininlärning (DVAAM, HT27)

Prerequisite QA of 53 course occurrences by 2 independent agent runs: `claude-code claude-sonnet-5-5`, `codex gpt-5.4`.

1. Executive summary

Two independent runs (Claude Code with `claude-sonnet-5-5`, Codex with `gpt-5.4`) agree on 49 of 53 verdicts, and all 14 priority issues were reported by both. The four high-severity issues are therefore well supported:

1. **C/C++ is assumed but never taught.** DV1697 Operativsystem (VT29) runs its labs mainly in C/C++, and DV2606 Multiprocessorprogrammering repeats the gap in all three offerings (HT29, HT30, HT31). The guaranteed programming path before them is Python (plus assembly in DV1730). Either add C/C++ to an earlier mandatory course or state it as a prerequisite.
2. **Two mandatory courses have no published course plan:** IY2658 (VT31) and DV2649, the 30 hp degree project (VT32). Their prerequisites cannot be checked. Four elective occurrences (SL2564, SL2565 twice, MA2517) are also missing plans.
3. **IY2650 (HT30 and HT31) depends on the elective IY2649.** A student who did not choose IY2649 does not meet the prerequisite.
4. **Imprecise prerequisite wording** in PA1484 and DV1688 (generic phrases that match no course code); both runs flagged it.

The runs disagree on four courses. For PA1484, PA1485 and DV1688 the Claude run gave UNCLEAR where Codex gave OK, so the difference is how strictly vague wording is judged, not whether a problem exists. For DV2606 (HT29) the Claude run gave CONDITIONAL and Codex gave GAP, so a human should decide it. Four low-severity findings come from the Claude run only (PA1481, DV1582, PA1485 version-control tooling, DV2658 placed before DV2646); verify these before acting. Agreement between two models is not proof; the findings are candidates for human review.

Key figures (computed):

- 2 runs agree on the verdict for 49 of 53 courses (92%); 4 courses have differing verdicts.
- 11 courses are flagged (not OK) by every run: DV1697 VT29, SL2564 HT30, DV2606 HT30, SL2565 HT30, IY2650 HT30, IY2658 VT31, MA2517 VT31, DV2606 HT31, IY2650 HT31, SL2565 HT31, DV2649 VT32.
- 14 issues were reported by more than half of the runs (14 by all); 4 by a minority of runs only and need verification.
- High-severity issues with majority or consensus support: 4.

2. Priority issues (reported by a majority of runs)

Agents describe the same problem in different words, so issues are grouped by course and finding type. *Support* = number of runs reporting it. *Severity* is the highest any run gave.

ID	Severity	Support	Course	Semester	Type	What	Recommendation
C-001	high	2/2	DV1697	VT29	competence-gap	DV1697 labs are done mainly in C/C++ and students are expected to know the language, but the mandatory programming courses before VT29 (DV1574, DV1582) use Python and DV1730 uses assembly. No mandatory course introduces C/C++.	Add a C/C++ introduction to an earlier mandatory course (e.g. DV1730 or DV1682), or state C/C++ as an explicit prerequisite and include a language primer at the start of DV1697.
C-002	high	2/2	DV2606	HT29	competenc	DV2606 expects prior C/C++ competence for the	Add C/C++ earlier in the mandatory path, redesign the

ID	Severity	Support	Course	Semester	Type	What	Recommendation
					e-gap	practical laboratory work, but the guaranteed mandatory path provides programming mainly in Python and does not explicitly teach C/C++ before the course.	course labs to use already taught languages, or state C/C++ explicitly as a prerequisite and ensure the program supplies it before DV2606.
C-003	high	2/2	IY2658	VT31	missing-course-plan	No published course plan for IY2658 (VT31, mandatory, 6 hp, Avancerad nivå); the fetch found no link on the program page.	Publish the course plan (with prerequisites, content and learning outcomes) before the program is offered, or remove the course from the program.
C-004	high	2/2	DV2649	VT32	missing-course-plan	No published course plan for DV2649 (VT32, mandatory, 30 hp, Avancerad nivå, degree project); the fetch found no link on the program page.	Publish the course plan (with prerequisites, content and learning outcomes) before the program is offered, or remove the course from the program.
C-005	medium	2/2	PA1484	HT28	ambiguous-requirement	The prerequisite ‘introduktion till ingenjörsarbete/-rollen’ (6 hp) does not match any course in the program by name or learning outcomes; DV1683 and PA1481 only partly touch the topic.	Name the intended course code(s) in the prerequisite, or reword it (e.g. to state ‘DV1683’) so that satisfaction can be verified; alternatively add an explicit engineering-role outcome to the intended course.
C-006	medium	2/2	DV1688	VT30	ambiguous-requirement	The prerequisite ‘datorsäkerhet och ingenjörsarbete, 5 hp’ does not correspond to any course in the program; DV1684 Säkerhetsteknik is only a partial match.	Replace the generic wording with the actual course code(s) (probably DV1684) in the course plan, or add DV1684 to an explicit equivalence list.
C-007	medium	2/2	SL2564	HT30	missing-course-plan	No published course plan for SL2564 (HT30, elective, 6 hp, Avancerad nivå); the fetch found no link on the program page.	Publish the course plan (with prerequisites, content and learning outcomes) before the program is offered, or remove the course from the program.
C-008	medium	2/2	DV2606	HT30	competence-gap	The labs require C/C++, but no earlier course teaches C or C++; DV1697 only uses it in labs while presupposing it.	Introduce C/C++ explicitly in a mandatory earlier course (see DV1697) or state C/C++ as a prerequisite and offer a primer here.
C-009	medium	2/2	SL2565	HT30	missing-course-plan	No published course plan for SL2565 (HT30, elective, 6 hp, Avancerad nivå); the fetch found no link on the program page.	Publish the course plan (with prerequisites, content and learning outcomes) before the program is offered, or remove the course from the program.
C-010	medium	2/2	IY2650	HT30	elective-dependency	IY2650 requires the elective IY2649 (or ‘motsvarande’); a student who has not chosen IY2649 does not meet the prerequisite through mandatory courses.	State in the program plan that IY2649 and IY2650 must be chosen together (in this order), or clarify what ‘motsvarande’ covers.
C-011	medium	2/2	MA2517	VT31	missing-course-plan	No published course plan for MA2517 (VT31, elective, 6 hp, Avancerad nivå); the fetch found no link on the program page.	Publish the course plan (with prerequisites, content and learning outcomes) before the program is offered, or remove the course from the program.
C-012	medium	2/2	DV2606	HT31	competence-gap	The labs require C/C++, but no earlier course teaches C or C++; DV1697 only uses it in labs while presupposing it.	Introduce C/C++ explicitly in a mandatory earlier course (see DV1697) or state C/C++ as a prerequisite and offer a primer here.

ID	Severity	Support	Course	Semester	Type	What	Recommendation
C-013	medium	2/2	IY2650	HT31	elective-dependency	This is a repeated offering of IY2650 (see seq 41). IY2650 requires the elective IY2649 (or ‘motsvarande’); a student who has not chosen IY2649 does not meet the prerequisite through mandatory courses.	State in the program plan that IY2649 and IY2650 must be chosen together (in this order), or clarify what ‘motsvarande’ covers.
C-014	medium	2/2	SL2565	HT31	missing-course-plan	No published course plan for SL2565 (HT31, elective, 6 hp, Avancerad nivå); the fetch found no link on the program page.	Publish the course plan (with prerequisites, content and learning outcomes) before the program is offered, or remove the course from the program.

3. Disagreements to adjudicate

Courses where the runs reached different verdicts. A human (or a further run) should decide these; reasoning of each run is quoted.

#	Semester	Course	Type	claude-code claude-sonnet-5-5	codex gpt-5.4
12	HT28	PA1484 Programvaruutveckling	mandatory	UNCLEAR	OK
23	HT29	PA1485 Mjukvaruutvecklingsprojekt i team	mandatory	UNCLEAR	OK
25	HT29	DV2606 Multiprocessorprogrammering	elective	CONDITIONAL	GAP
28	VT30	DV1688 Digital undersökningsteknik	elective	UNCLEAR	OK

12. PA1484 Programvaruutveckling (HT28)

- **claude-code claude-sonnet-5-5**: UNCLEAR (medium). PA1484 requires ‘6 avklarade hp från kurser i introduktion till ingenjörsarbete/-rollen och minst 6 avklarade hp i programmering’. Programming is clearly met by DV1574 and DV1582 (12 hp). No earlier course is titled or scoped as an introduction to the engineering role: DV1683 only ‘introducerar begreppet ingenjörsmässighet’ and PA1481 covers ‘Professionella praxis och riktlinjer för etik för ingenjörer’, neither wit…
- **codex gpt-5.4**: OK (medium). The plan requires “6 avklarade hp från kurser i introduktion till ingenjörsarbete/-rollen och minst 6 avklarade hp i programmering”. Before seq 12 the ledger guarantees at least 12 hp of programming through DV1574 and DV1582, and it also guarantees earlier orientation courses about AI, engineering professionalism, ethics and sustainability through DV1683 and PA1481. The progression therefore works, but the first pre…

23. PA1485 Mjukvaruutvecklingsprojekt i team (HT29)

- **claude-code claude-sonnet-5-5**: UNCLEAR (medium). PA1485 requires ‘12 avklarade hp i programmering och 6 avklarade hp i programvaruutveckling’: DV1574 + DV1582 give 12 hp programming and PA1484 Programvaruutveckling gives 6 hp software development, all finished before HT29. The course expects students to ‘Utveckla och testa en lösning … med hjälp av lämpliga utvecklingsmetoder och verktyg’ in a team; no course plan in the program mentions version control (git) an…
- **codex gpt-5.4**: OK (high). The formal prerequisites are “12 avklarade hp i programmering och 6 avklarade hp i programvaruutveckling”. Before seq 23 the ledger guarantees DV1574 and DV1582 for 12 hp of programming and PA1484 Programvaruutveckling for 6 hp in software development. That earlier progression directly supports the course’s team-based work on requirements, agile development, implementation and testing.

25. DV2606 Multiprocessorprogrammering (HT29)

- **claude-code claude-sonnet-5-5**: CONDITIONAL (medium). DV2606 (elective) requires ‘60 hp i huvudområdet datavetenskap eller programvaruteknik inklusive 12 hp programmering, 6 hp algoritmer och datastrukturer, operativsystem samt datorteknik 6 hp’. The ledger shows 72 hp Datavetenskap and 36 hp Programvaruteknik guaranteed; DV1574 + DV1582, DV1682, DV1697 and DV1730 are all mandatory and finished before HT29 period 2. Labs are ’huvudsakligen i programmeringsspråken C / C…

- **codex gpt-5.4**: GAP (high). Formally, the student satisfies the requirement of 60 hp in datavetenskap/programvaruteknik plus earlier study in programming, algorithms, operating systems and computer engineering. The ledger before seq 25 guarantees 72 hp in Datavetenskap and 36 hp in Programvaruteknik, together with DV1574/DV1582, DV1682, DV1697 and DV1730. However, the learning activities state that the lab assignments are carried out mainly in…

28. DV1688 Digital undersökningsteknik (VT30)

- **claude-code claude-sonnet-5-5**: UNCLEAR (medium). DV1688 (elective) requires '60 hp … i huvudområdet datavetenskap eller programvaruteknik inklusive datorsäkerhet och ingenjörsarbete, 5 hp, programmering, 5 hp, samt operativsystem, 5 hp'. The 60 hp, programming (DV1574, DV1582) and operating systems (DV1697) parts are met by mandatory courses. 'Datorsäkerhet och ingenjörsarbete' is not the name of any course in the program; DV1684 Säkerhetsteknik (6 hp) covers co…
- **codex gpt-5.4**: OK (medium). The plan requires 60 hp in Datavetenskap/Programvaruteknik including "datorsäkerhet och ingenjörsarbete, 5 hp, programmering, 5 hp, samt operativsystem, 5 hp". Before seq 28 the ledger guarantees well over 60 hp in the relevant main areas, programming through DV1574/DV1582, operating systems through DV1697, and security through DV1684; the student also has networking through DV1685, which is useful for the stated le…

4. Findings reported by a minority of runs

Not corroborated by a majority; treat as hypotheses. If another run flagged the same course under a different finding type, it is noted.

ID	Severity	Reported by	Course	Semester	Type	What	Other runs on this course
C-015	low	claude-code claude-sonnet-5-5	PA1481	VT28	ambiguous-requirement	The plan requires Fysik 2 and Kemi 1, which have no visible connection to the content and are stricter than other first-term courses (MA1498 requires Fysik 1 only). It is unclear whether the program's admission requirements guarantee Fysik 2 and Kemi 1.	-
C-016	low	claude-code claude-sonnet-5-5	DV1582	VT28	ambiguous-requirement	The requirement says 'genomgången kurs … 6 hp, varav minst 2 hp ska vara avklarade', which is self-contradictory (a completed course versus at least 2 of 6 hp completed).	-
C-017	low	claude-code claude-sonnet-5-5	PA1485	HT29	competence-gap	Team-based development needs collaborative tooling (version control), but none of the program's course plans mentions it, and PA1485's plan does not state that it is taught.	-
C-018	low	claude-code claude-sonnet-5-5	DV2658	HT30	ordering	DV2658 fine-tunes and develops neural generative models, but the dedicated deep learning course DV2646 Djup maskininlärning is placed in VT31, after DV2658 (HT30). Students rely on only introductory deep learning from DV2659/DV1683.	-

5. Runs and agreement

Harness	Model	Date	OK	CONDITIONAL	GAP	UNCLEAR	Findings (high/medium/low)
claude-code	claude-sonnet-5-5	2026-10-21	38	5	1	9	3/11/4
codex	gpt-5.4	2026-10-21	41	4	2	6	4/8/2

Pairwise agreement. *Exact* = same verdict; *flag* = both say OK or both say not OK.

Run A	Run B	Courses	Exact verdict	Flag (OK vs not OK)
claude-code claude-sonnet-5-5	codex gpt-5.4	53	49 (92%)	50 (94%)

6. Course matrix

#	Semester	Course	Type	claude-code claude-sonnet-5-5	codex gpt-5.4	Consensus
1	HT27	MA1498 Linjär algebra	mandatory	OK	OK	**OK**
2	HT27	DV1574 Programmering och problemlösning med Python	mandatory	OK	OK	**OK**
3	HT27	DV1683 Artificiell intelligens och ingenjörsmässighet	mandatory	OK	OK	**OK**
4	HT27	MA1499 Envariabelanalys 1: funktioner och differentialkalkyl	mandatory	OK	OK	**OK**
5	HT27	IY1458 Industriell ekonomi, översiktskurs	mandatory	OK	OK	**OK**
6	VT28	MA1500 Envariabelanalys 2: differentialekvationer och integralkalkyl	mandatory	OK	OK	**OK**
7	VT28	PA1481 Digital etik och hållbar utveckling	mandatory	OK	OK	**OK**
8	VT28	DV1582 Programmering i Python, fortsättningskurs	mandatory	OK	OK	**OK**
9	VT28	MA1504 Diskret matematik	mandatory	OK	OK	**OK**
10	VT28	DV1682 Datastrukturer och algoritmer	mandatory	OK	OK	**OK**
11	HT28	MA1514 Flervariabelanalys	mandatory	OK	OK	**OK**
12	HT28	PA1484 Programvaruutveckling	mandatory	UNCLEAR	OK	**SPLIT** (not unanimous)
13	HT28	FY1436 Kraft och energi	mandatory	OK	OK	**OK**
14	HT28	DV1730 Datorteknik	mandatory	OK	OK	**OK**
15	HT28	MS1419 Matematisk statistik	mandatory	OK	OK	**OK**
16	VT29	DV1697 Operativsystem	mandatory	GAP	GAP	**GAP**
17	VT29	DV1663 Databasteknik	mandatory	OK	OK	**OK**
18	VT29	DV1685 Datakommunikation och nätverksteknik	mandatory	OK	OK	**OK**

#	Semester	Course	Type	claude-code claude-sonnet-5-5	codex gpt-5.4	Consensus
19	VT29	DV1684 Säkerhetsteknik	mandatory	OK	OK	**OK**
20	VT29	DV1597 Intelligent dataanalys	mandatory	OK	OK	**OK**
21	HT29	ET1558 Dynamiska system och återkoppling	mandatory	OK	OK	**OK**
22	HT29	DV2659 Tillämpad artificiell intelligens	mandatory	OK	OK	**OK**
23	HT29	PA1485 Mjukvaruutvecklingsprojekt i team	mandatory	UNCLEAR	OK	**SPLIT** (not unanimous)
24	HT29	PA1486 Testning av mjukvara	elective	OK	OK	**OK**
25	HT29	DV2606 Multiprocessorprogrammering	elective	CONDITIONAL	GAP	**SPLIT** (not unanimous)
26	HT29	DV2638 Maskininlärning	mandatory	OK	OK	**OK**
27	VT30	DV1601 Användbarhet och interaktionsdesign	elective	OK	OK	**OK**
28	VT30	DV1688 Digital undersökningsteknik	elective	UNCLEAR	OK	**SPLIT** (not unanimous)
29	VT30	TE1429 Kandidatarbete i teknik för AI och maskininlärning	mandatory	OK	OK	**OK**
30	VT30	DV2647 Penetrationstestning och etisk hackning	elective	OK	OK	**OK**
31	VT30	PA2601 DevSecOps - Utveckling, säkerhet och drift	elective	OK	OK	**OK**
32	HT30	DV2583 Digital etik	mandatory	OK	OK	**OK**
33	HT30	IY2649 Technology venture 1: Innovation och affärsutveckling	elective	OK	OK	**OK**
34	HT30	DV2607 Säkerhet i AI-system	mandatory	OK	OK	**OK**
35	HT30	SL2564 Förståelse för marknader, företagande och design inom hållbarhetsomställningar	elective	UNCLEAR	UNCLEAR	**UNCLEAR**
36	HT30	MA1513 Kryptologins grunder	elective	OK	OK	**OK**
37	HT30	DV2606 Multiprocessorprogrammering	elective	CONDITIONAL	CONDITIONAL	**CONDITIONAL**
38	HT30	SL2565 Bedöma och designa inom	elective	UNCLEAR	UNCLEAR	**UNCLEAR**

#	Semester	Course	Type	claude-code claude-sonnet-5-5	codex gpt-5.4	Consensus
		omställningar till hållbarhet				
39	HT30	DV2658 Generativ artificiell intelligens	elective	OK	OK	**OK**
40	HT30	MS2506 Bayesiansk statistik	mandatory	OK	OK	**OK**
41	HT30	IY2650 Technology venture 2: kommersialisering av innovationer	elective	CONDITIONAL	CONDITIONAL	**CONDITIONAL**
42	VT31	IY2658 ICT start-ups och högteknologiskt entreprenörskap	mandatory	UNCLEAR	UNCLEAR	**UNCLEAR**
43	VT31	MS1415 Robusta metoder	mandatory	OK	OK	**OK**
44	VT31	DV2640 Avancerad maskininlärning	mandatory	OK	OK	**OK**
45	VT31	ET2627 Robotik	elective	OK	OK	**OK**
46	VT31	MA2517 Kvantinformation och algoritmer för kvantdatorer	elective	UNCLEAR	UNCLEAR	**UNCLEAR**
47	VT31	DV2646 Djup maskininlärning	mandatory	OK	OK	**OK**
48	HT31	PA2611 Avancerat mjukvaruutvecklingsprojekt i team	mandatory	OK	OK	**OK**
49	HT31	DV2654 Forskningsmetodik i datavetenskap	mandatory	OK	OK	**OK**
50	HT31	DV2606 Multiprocessorprogrammering	elective	CONDITIONAL	CONDITIONAL	**CONDITIONAL**
51	HT31	IY2650 Technology venture 2: kommersialisering av innovationer	elective	CONDITIONAL	CONDITIONAL	**CONDITIONAL**
52	HT31	SL2565 Bedöma och designa inom omställningar till hållbarhet	elective	UNCLEAR	UNCLEAR	**UNCLEAR**
53	VT32	DV2649 Masterarbete i datavetenskap	mandatory	UNCLEAR	UNCLEAR	**UNCLEAR**

7. Method and limits

- Source: the validated per-run reports (`DVAAM-HT27__claude-code__claude-sonnet-5-5__2026-10-21.json`, `DVAAM-HT27__codex__gpt-5.4__2026-10-21.json`). Each run reviewed the same program independently; this report only compares them.
- Consensus verdict = the verdict given by more than half of the runs, otherwise SPLIT. Issues are matched by course and finding type, so one problem reported under two types counts as two minority issues; look at the “other runs” column before dismissing a minority finding.

- Agreement between runs is not correctness: models can share the same blind spot. Majority support raises confidence, it does not prove the finding. All verdicts are LLM judgments of course-plan text and flag candidates for human review.
- The narrative in section 1 was written by an agent from the figures in this report; everything else is computed.